

Map Representation

Module 4

Map Representation in Robotics

- When designing a map for a robot, we need to consider how the environment is represented.
- This directly affects how we track the robot's position.
- The accuracy of the map influences how precisely we can determine where the robot is.

Map Representation in Robotics

There are three key factors to consider when choosing a map representation:

1. **Accuracy vs. Goals** – The map should be as precise as needed for the robot to complete its tasks.
2. **Map vs. Sensor Data** – The level of detail in the map should match the type and accuracy of the robot's sensors.
3. **Complexity vs. Computation** – A more detailed map requires more computing power for tasks like mapping, localization, and navigation.
 - Different types of maps come with various trade-offs.
 - Choosing the right map depends on understanding these trade-offs and considering the specific needs of the robot's environment and tasks.

Continuous Map Representations in Robotics

- A **continuous map** is a way to represent the environment with precise positions of objects in continuous space.
- These maps are usually used in **2D** because adding more dimensions makes computations much more complex.
- To balance accuracy and efficiency, continuous maps often assume that they include all relevant objects and that empty areas in the map correspond to empty spaces in the real world.
- This reduces memory usage because only areas with objects need to be stored.

Continuous Map Representations in Robotics

How Continuous Maps Work

- One common method is to use **polygons** to represent obstacles.
- For example, some mobile robot path-planning systems store only the necessary geometric shapes to define obstacles without including unnecessary details like texture or color.
- This makes calculations faster and reduces memory usage.
- A good example is **line extraction** from laser rangefinders.
- Many indoor robots use laser sensors to measure distances to nearby objects.
- These sensors collect thousands of data points, and the system simplifies them into straight lines to represent walls and corners.
- This keeps the map relevant to the robot's sensors and reduces unnecessary details.

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Advantages of Continuous Maps

1. **High Accuracy** – The robot's position can be defined very precisely in a continuous coordinate space.
2. **Flexible Position Representation** – The robot's possible position can be shown as a shape (like an oval) or as a set of sampled points around its most likely location.
3. **Efficient Storage with Abstraction** – By focusing only on key features, such as walls and corners, continuous maps can sometimes require **less** memory than traditional discrete maps.

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Challenges of Continuous Maps

- The main challenge is that continuous maps can require **high computational power**.
- However, this can be managed by simplifying the map and storing only essential details.
- By using techniques like **abstraction** and the **closed-world assumption** (assuming all necessary objects are included in the map), continuous maps can sometimes be just as efficient as simpler discrete maps.

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Continuous Map Representations in Robotics

- An example of this type of map, shown in Figure is a 2D model where obstacles are represented as polygons in a continuous coordinate space.
- This approach is similar to the one used by Latombe [21, 98] and others for mapping environments in mobile robot path planning.

